



Chemistry Ch 4.9: Metals and Non-Metals Compared

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Property	Metals	Non-metals
Thermal conductivity	Ache conductor	Khraab (insulators)
Electrical conductivity	Ache conductor	Khraab (graphite istisna)
Malleability	Peet kar chaadar ban jati hai	Bhurbhure (brittle), toot jate hain
Ductility	Taar khinch jati hai	Nahin khinchti
Melting / boiling point	Aam tor par ooncha	Aam tor par neecha
Ion banana	Electron khote hain → cation	Electron lete hain → anion
Table mein jagah	Bayein taraf	Dayein taraf

Saari table ek hi baat se nikalti hai

Dhat mein electrons **azad** hote hain, poore maddey mein ghoomte hue, aur isi ko **sea of electrons** kehte hain.

Azad electrons **bijli** le jate hain, aur **garmi** bhi. Aur jab aap dhat ko peetein to atoms ki tehen **phisal** jati hain magar electron ka samundar unhein jore rakhta hai, is liye wo **tootti nahin, phail** jati hai. Yehi malleability aur ductility hai.

Non-metal mein electrons bandhay hue hain, is liye na bijli chalti hai aur na wo phailta hai, bas **toot** jata hai.

Do istisna yaad rakhein

Graphite ek non-metal hai magar **bijli chalata** hai, kyunke us mein chautha electron azad hai.

Mercury ek dhat hai magar kamre ke temperature par **maaye** hai.

Quiz

1. Compare metals and non-metals for thermal and electrical conductivity, malleability and ductility, and melting point. Say which side of the periodic table each is on.
2. Explain, using the idea of free electrons, why metals conduct electricity and can be hammered into sheets while non-metals are brittle. Give one exception to each rule.